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HIGH-FREQUENCY DATA AND LIMIT ORDER BOOKS  
A FEW EXERCISES ON POISSON PROCESSES

**Exercise 1** (Poisson characterization with stationary and independent increments).

Let  $(N_t)_{t \geq 0}$  be a counting process such that the following conditions are satisfied :

- (i)  $N$  has independent increments ;
- (ii)  $N$  has stationary increments ;
- (iii) for any interval  $I$  with finite length,  $N(I)$  is integrable ;
- (iv) there exists an interval  $I$  such that  $\mathbf{P}(N(I) > 0) > 0$ .
- (v) the process is simple :  $\lim_{\epsilon \rightarrow 0} \frac{\mathbf{P}(N(\epsilon) \geq 2)}{\epsilon} = 0$ .

The aim of the exercise is to show that the number of points in any interval is Poisson distributed. Some standard results are recalled along the way.

1. We start by showing that the expectation of  $N_t$  is linear with respect to  $t$ .
  - (a) Let  $m(t) = \mathbf{E}[N_t]$ ,  $t \geq 0$ . Prove that for all  $s, t \geq 0$ ,  $m(t+s) = m(t) + m(s)$ .
  - (b) Prove that there exists  $\lambda > 0$  such that  $m(t) = \lambda t$ .
2. Let  $(p_n)_{n \in \mathbb{N}}$  be a sequence with values in  $[0, 1]$  such that  $\lim_{n \rightarrow +\infty} np_n = \mu > 0$ . Let  $Z_n$  be a r.v. with binomial distribution with parameters  $n, p_n$ . Prove that for all  $k \in \mathbb{N}$ ,  
$$\lim_{n \rightarrow +\infty} \mathbf{P}(Z_n = k) = e^{-\mu} \frac{\mu^k}{k!}.$$
3. Let  $t > 0$ . For all  $n \in \mathbb{N}^*$ , let  $t_j = \frac{jt}{n}$  and  $X_j^{(n)} = N_{t_j} - N_{t_{j-1}}$ ,  $j = 1, \dots, n$ . Let also  
$$Y_j^{(n)} = \min(X_j^{(n)}, 1), j = 1, \dots, n, \text{ and } N_t^{(n)} = \sum_{j=1}^n Y_j^{(n)}.$$
  - (a) Show that the random variables  $Y_j^{(n)}$ ,  $j = 1, \dots, n$  are i.i.d. with Bernoulli distribution. Let  $p_n = \mathbf{P}(Y_1^{(n)} = 1)$ .
  - (b) Prove that  $\lim_{n \rightarrow +\infty} \mathbf{P}(N_t^{(n)} \neq N_t) = 0$ .
  - (c) Prove that for all  $k \in \mathbb{N}$ ,  $\left| \mathbf{P}(N_t = k) - \mathbf{P}(N_t^{(n)} = k) \right| \leq \mathbf{P}(N_t^{(n)} \neq N_t)$ .
4. Let  $Z$  be an integer-valued random variable. Prove that  $\mathbf{E}[Z] = \sum_{l=1}^{\infty} \mathbf{P}(Z \geq l)$ .
5. Prove that  $\lim_{n \rightarrow \infty} \mathbf{E}[N_t^{(n)}] = \mathbf{E}[N_t]$ .
6. Prove that  $N_t$  has a Poisson distribution with parameter  $\lambda t$ .

**Exercise 2** (Superposition of independent Poisson processes).

Let  $(N_t^{(1)})_{t \geq 0}$  and  $(N_t^{(2)})_{t \geq 0}$  be two independent Poisson processes with intensities  $\lambda_1 > 0$  and  $\lambda_2 > 0$ . Let  $N_t = N_t^{(1)} + N_t^{(2)}$ . Let  $\lambda = \lambda_1 + \lambda_2$ .

1. Apply the superposition theorem of random Poisson measures to this specific case.  
The remainder of the exercise is a direct proof of the result, ignoring the general superposition theorem.
2. Prove that for all  $0 \leq s < t$ , and all  $k \in \mathbb{N}$ ,  $\mathbf{P}(N_t - N_s = k) = e^{-\lambda(t-s)} \frac{\lambda^k (t-s)^k}{k!}$ .
3. Prove that for all  $0 \leq s < t \leq u < v$ ,  $N_t - N_s$  and  $N_v - N_u$  are independent.
4. Conclude.

**Exercise 3** (Marked Poisson process).

Let  $(T_n)_{n \geq 1}$  be the event times of a time-homogeneous Poisson process  $N$  with parameter  $\lambda > 0$ . Let  $(X_n)_{n \geq 1}$  be a sequence of i.i.d. random variables with Bernoulli distribution with parameter  $p \in [0, 1]$ , independent of  $(T_n)$ . The process  $\{(T_n, X_n), n \geq 1\}$  is a marked Poisson process (each point  $T_n$  is given a mark  $X_n$ ).

Let  $\mathcal{T}^{(i)} = \{T_n : X_n = i\}$ ,  $i = 0, 1$ , be the subset of times for each mark. Let  $(N_t^{(0)})_{t \geq 0}$  and  $(N_t^{(1)})_{t \geq 0}$  be the counting processes associated to these subset of points.

1. Apply the marking theorem of random Poisson measures to this specific case.  
The remainder of the exercise is a direct proof of the result, ignoring the general marking theorem.
2. Compute  $\mathbf{P}(N_t^{(1)} - N_s^{(1)} = k)$  and  $\mathbf{P}(N_t^{(0)} - N_s^{(0)} = k)$  for all  $0 \leq s < t$ , and all  $k \in \mathbb{N}$ .
3. Prove that for all  $0 \leq s < t \leq u < v$ ,  $N_t^{(0)} - N_s^{(0)}$ ,  $N_t^{(1)} - N_s^{(1)}$ ,  $N_v^{(0)} - N_u^{(0)}$  and  $N_v^{(1)} - N_u^{(1)}$  are independent.
4. Conclude.

**Exercise 4** (Some conditional distributions).

Let  $N$  be a time-homogeneous Poisson process with parameter  $\lambda > 0$ . Let  $(T_n)_{n \geq 1}$  be the sequence of associated events.

1. Compute the conditional distribution of  $N_s$  given  $N_t = l$ , for all  $0 \leq s \leq t$ ,  $l \in \mathbb{N}$ .
2. Compute the conditional distribution of  $T_1$  given  $N_t = 1$ .
3. Check that the previous result is in agreement with the more general result on Poisson random measures proved in class.
4. Compute the conditional distribution of  $(T_1, \dots, T_n)$  given  $N_t = n$ .
5. Check that the previous result is in agreement with the more general result on Poisson random measures proved in class.

**Exercise 5** (Laplace functional of a Poisson random measure).

Let  $N$  be a point process on  $(E, \mathcal{B}(E))$ . For any non-negative measurable function  $f : E \rightarrow [0, +\infty)$ , let us define

$$L_N(f) = \mathbf{E} \left[ \exp \left( - \int_E f(x) N(dx) \right) \right].$$

$L_N$  is called the *Laplace functional* of  $N$ .

1. Let  $X$  be a Poisson r.v. with parameter  $\lambda$ . Compute  $\mathbf{E}[e^{-tX}]$ ,  $t \geq 0$ .

2. Let  $N$  be a Poisson random measure with intensity measure  $\mu$ . Prove that, for any non-negative measurable function  $f : E \rightarrow [0, +\infty)$ ,

$$L_N(f) = \exp \left( - \int_E (1 - e^{-f(x)}) \mu(dx) \right).$$

**Exercise 6** (Covariance of increments).

Let  $N$  be a Poisson random measure on  $(E, \mathcal{B}(E))$  with intensity measure  $\mu$ . Let  $A, B \in \mathcal{B}(E)$  such that  $\mu(A) < \infty$  and  $\mu(B) < \infty$ . Show that  $\text{Cov}(N(A), N(B)) = \mu(A \cap B)$ .

**Exercise 7** (A family of martingales).

Let  $N$  be a time-homogeneous Poisson process with parameter  $\lambda > 0$ . Let  $\alpha \in \mathbb{R}$  and  $Y_t^{(\alpha)} = \exp(\alpha N_t - \lambda t(e^\alpha - 1))$ . Prove that  $(Y_t^{(\alpha)})_{t \geq 0}$  is a martingale (w.r.t. the filtration  $(\mathcal{F}_t)_{t \geq 0}$  generated by the Poisson process).